PPR Land Acquisition Under Climate Non-Stationarity

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Overview

This project asks whether accounting for projected future ecological conditions (non-stationarity) leads to improved land-acquisition decisions for waterfowl conservation in the U.S. Prairie Pothole Region (PPR) than assuming stationarity. It builds three decision models and compares their performance across climate-change scenarios and a stationary null.

The three models acquire land period by period (decadal steps, 2020–2100) under a budget:

- **Greedy** — status-quo heuristic: rank parcels by current benefit-to-cost ratio, buy down the list. No foresight.
- **Myopic ILP** — re-optimizes each decade but freezes current benefit and hazard and assumes they persist (a fabricated stationary future); implements the current decade, then re-solves.
- **Rolling-horizon ILP** — same structure, but uses the true projected trajectories. The foresighted benchmark.

The spatial unit is the **EAU** (Equal Area Unit, ~342 km^2). Roughly 1,200 EAUs are gridded across the 24 Wetland Management Districts (WMDs), and 1,169 of them have FOREsce land-cover coverage. Four WMDs are then dropped in script `04`: Benton Lake, Bowdoin and Northeast Montana (low confidence in hydrological modeling) and Fergus Falls (insufficient land-cover coverage). The analysis therefore uses **879 EAUs across 20 WMDs**, over **9 decadal periods** and **5 scenarios** (RCP 4.5/8.5 × wet/dry, plus a stationary null).

The main results use the FOREsce GCAM Reference land-cover scenario, a budget of 5 median-priced EAUs per period, and the conversion hazard derived from the land-cover projections. Three sensitivity analyses are built into the workflow:

- **Land cover** — an alternative land-cover scenario (FOREsce BAU).
- **Budget** — 2, 5 and 10 median-priced EAUs per period.
- **Conversion hazard** — the hazard rescaled to a series of published annual conversion rates (0.28, 0.40, 0.57, 0.95 and 1.33%/yr), with the canonical run as the lowest point.

Requirements

- **R** with the following packages, checked by `00_setup.R`:
  - core (required): `dplyr`, `tidyr`, `readr`, `stringr`, `purrr`, `ggplot2`, `scales`, `Matrix`, `jsonlite`;
  - spatial: `terra`, `sf` (required by `01`, `02`, `06`; optional for `10`, where they are used only for the WMD outline overlay on maps);
  - reporting: `flextable`, `knitr` (used by `11_results.qmd`).
- The native **Gurobi** R package with an active (academic) license, for `07`–`09`.
- **Quarto** to render `11_results.qmd`.

How to run

1. **Main run** — `Rscript 00_run_all.R` runs the data pipeline (`01`–`06`), the models (`08`) and the reporting assets (`10`) for the land-cover scenario set in `00_config.R` (normally `gcam_ref`). Outputs go to `runs/<LC_SCENARIO>/`. Leave the `PPR_BUDGET` and `PPR_HAZARD` environment variables unset for this run.

- Part of the pipeline can be run with `Rscript 00_run_all.R 08` (from `08` onward) or `Rscript 00_run_all.R 03 06` (a range).
- `02` takes about 60–75 min.
- `09_validation.R` certifies the formulation (P1–P5). It is skipped by default; set `RUN_VALIDATION <- TRUE` in `00_run_all.R` (or run it directly) when the formulation changes. It does *not* need to be re-run every time `08` is run.

2. **Budget sensitivity** (optional) — `Rscript 13_run_budget_sweep.R` re-runs `08` and `10` at budgets of 2, 5 and 10 median-priced EAUs per period, reusing the derived data from step 1 (land cover is not reprocessed). Each budget gets its own folder, `runs/<LC_SCENARIO>_bXX/`. Then `Rscript 10_results_assets.R` (with `PPR_BUDGET` unset) builds the cross-budget tables and figures. The budget-5 run reproduces the main run, and the sweep checks this. ~30–45 min per budget.
3. **Land-cover comparison** (optional) — set `LC_SCENARIO <- "bau"` in `00_config.R`, run `Rscript 00_run_all.R`, set it back to `"gcam_ref"`, then run `Rscript 10_results_assets.R`.
4. **Conversion-hazard sensitivity** (optional) — after step 1, `Rscript 15_run_hazard_sweep.R` runs the hazard sweep. For each target annual conversion rate it builds a rescaled panel (`14`) and then runs `08` and `10`, in its own folder, `runs/<LC_SCENARIO>_haz<tag>/` (e.g. `_haz1p33` for 1.33%/yr). Land cover is not reprocessed. Then, with `PPR_HAZARD` unset, `Rscript 10_results_assets.R` builds the hazard-sweep assets.
   - A single rate can be run by hand with `PPR_HAZARD=1.33 Rscript 14_build_hazard_panel.R` followed by `PPR_HAZARD=1.33 Rscript 00_run_all.R 08`. Optionally run `PPR_HAZARD=1.33 Rscript 09_validation.R` once.
   - `PPR_HAZARD` and `PPR_BUDGET` cannot be combined; hazard runs use the canonical budget.
5. **Results document** — with `PPR_BUDGET` and `PPR_HAZARD` unset, run `quarto render 11_results.qmd`.

Input data

`input_data/` is not distributed with the repository. Assemble it as follows:

Path	Source
`input_data/Wetland_Management_Districts/FSMS_WMD.shp` (with `.dbf`, `.prj`, `.shx`, and other sidecar files)	USFWS Wetland Management District boundaries
`input_data/prairie_potholes_gcam_ref_meanclim/` — only `prairie_potholes_gcam_ref_meanclim_2020.tif` is read	USGS FOREsce, GCAM Reference: https://www.sciencebase.gov/catalog/item/5b683db2e4b006a11f75b06a
`input_data/prairie_potholes_gcam_ref_rcp45/` — 10 GeoTIFFs, 2014–2100	as above, RCP 4.5
`input_data/prairie_potholes_gcam_ref_rcp85/` — 10 GeoTIFFs, 2014–2100	as above, RCP 8.5
`input_data/prairie_potholes_bau_{meanclim,rcp45,rcp85}/` — same layout as above	USGS FOREsce, BAU (only needed for the land-cover comparison)
`input_data/PADUS4_1VectorAnalysis_PADUS_Only/PADUS4_1VectorAnalysis_PADUS_Only.gdb`	USGS GAP PAD-US 4.1 Vector Analysis: https://doi.org/10.5066/P96WBCHS
`input_data/benefit.csv`	Waterfowl abundance projections (McKenna et al. 2026)
`input_data/places_fmv_pnas_dryad/1 estimates/places_fmv_vacant.tif`	Nolte 2020 PLACES FMV: https://datadryad.org/dataset/doi:10.5061/dryad.np5hqbzq9

Scripts

Setup and drivers:

- `00_config` — single source of run settings: land-cover scenario (`LC_SCENARIO`), budget (`BUDGET_EAUS_PER_PERIOD`, canonical 5; set from the `PPR_BUDGET` environment variable for budget runs), conversion-hazard target (`PPR_HAZARD`, for hazard runs), run IDs and folder paths.
- `00_setup` — loads packages, sources `00_config`, and creates the run folders.
- `00_run_all` — runs the pipeline in order (see How to run), each script in a fresh R session. Refuses to re-run `01`–`06` for a budget or hazard run.
- `13_run_budget_sweep` — budget sensitivity driver: runs `08` and `10` for each budget, skips budgets already complete, logs each run to `logs/budget_sweep/`, and checks the budget-5 run against the main run.
- `15_run_hazard_sweep` — conversion-hazard sensitivity driver: for each rate in `RATES` (0.28, 0.40, 0.57, 0.95, 1.33%/yr, each tied to a published estimate), builds the rescaled panel with `14`, then runs `08` and `10`.

Data pipeline:

1. `01_create_EAUs` — grids the WMD shapefile into EAUs and builds an EAU↔WMD lookup. Writes `wmd_raster_equal_area.tif`, `eau_wmd_lookup.{rds,csv}` and `wmd_summary.csv` to `runs/common/`, shared by all runs.
2. `02_EAU_prop_suitable` — proportion of suitable habitat per EAU from USGS FOREsce land-cover projections, over time and by RCP (resampling section, ~8 min), plus gross habitat loss on unprotected land (flow pass at native 30 m resolution, ~60–75 min). The PAD-US 4.1 protected-area mask (GAP 1–3) is applied only to the gross-loss calculation used for the conversion hazard; benefit allocation within WMDs uses the unmasked land cover.
3. `03_create_data_panel` — assembles the long-format data panel (EAU × year × RCP × GCM), including wet/dry duplication and the stationary scenario. Land cover is joined to EAUs by raster cell (`cell_id`), not by row order. Also translates the gross flows from `02` to EAUs (`eau_flows.rds`).
4. `04_benefit_data` — waterfowl-abundance benefit, interpolated to decadal steps (benefit windows anchored at their midpoints: 2012, 2045, 2085) and distributed within WMDs by suitable habitat. **Drops the 4 excluded WMDs** (→ 879 EAUs).
5. `05_risk_of_loss` — conversion-risk hazard: gross suitable habitat loss on unprotected habitat (i.e., a habitat erosion proxy), with a background floor ε. By default ε is one tenth of the smallest positive loss in the data; it can be set by hand with `epsilon_override`.
6. `06_cost_data` — acquisition cost per EAU (Nolte 2020 FMV, vacant land; 2%/yr inflation from 2017 dollars). Writes the final `data_panel.rds`.

Modeling core:

7. `07_ilp_core` — shared engine sourced by `08` and `09`: scenario-matrix assembly, survival/value/budget functions, the Gurobi solver wrapper, the three policies, and evaluation. Solver settings (a 60 s per-solve time cap, threads, seed, output) and the **spend-down** block live in parameter blocks near the top.
   - Temporal discount factor lives here: defined as 0.95 per decade.
   - Spend down mode lives here: budget deployment is controlled by `SPEND_DOWN_MODE` (default `"spend"` forces the model to exhaust the budget each time period).
   - Budget per period = `BUDGET_EAUS_PER_PERIOD` × the median EAU cost in that period (`make_budget()`). The number of EAUs is set in `00_config`, not here.
8. `08_run_models` — driver: runs the three models on all scenarios, scores each against its true future, and writes results, trajectories, a convergence summary, and the **per-parcel acquisition schedule joined to EAU coordinates** (`acquisition_schedule_spatial.csv`, also embedded in `model_results.rds`). The schedule join reads `runs/common/eau_wmd_lookup.rds`; if absent it warns and writes the schedule without coordinates rather than failing.
   - `REPRODUCIBLE <- TRUE` is the default: the solver runs single-threaded and seeded, for bit-reproducible reported numbers *and* a stable map footprint. Set it to `FALSE` for faster multi-threaded runs.
9. `09_validation` — correctness suite (P1–P5). Formulation tests run with spend-down off; the ordering test runs in the active mode.

Reporting / mapping:

10. `10_results_assets` — single pure-reader reporting script: builds results tables (as tidy CSVs of raw numbers), figures and maps from `08`'s saved outputs into `output_figs/`. Figures are saved as PNG and PDF. No solver/`07` needed. The maps need only `ggplot2`; `terra`/`sf` are used solely for the optional WMD outline overlay (auto-skipped if absent). Toggle maps with `DRAW_MAPS`. Per-run assets:

    - *Tables*: `table_rolling_improvement.csv`, `table_over_baseline.csv`.
    - *Figures*: `overtime_gap`, `decline`.
    - *Descriptive summary of model inputs* (read from the data panel, so they render even before any model has run): `fig_benefit_ribbon`, `map_benefit_2020_2100`, `map_cost_binned`, `fig_risk_box`.
    - *Acquisition maps*: `foresight_difference`, `prevented_loss_diff`.
    - *Metadata*: `run_metadata.json` (generation date and land-cover label, used to stamp the results document).

    At the end, three comparison sections write into the main run's `output_figs/`. Each runs only in the canonical configuration (`PPR_HAZARD` and `PPR_BUDGET` unset) and only when its input runs exist; otherwise it is skipped with a message, so `10` still runs normally inside every budget and hazard run. Rerun `Rscript 10_results_assets.R` after new comparison runs to refresh them.

    - *Land-cover comparison* — this run vs the alternative land-cover run (`COMPARE_RUN`, normally `bau`): `cmp_headline.csv`, `cmp_value_added.csv`, `cmp_risk_table.csv`, `fig_cmp_risk_box`, `cmp_metadata.json`.
    - *Budget sensitivity* — every complete `runs/<LC_SCENARIO>_bXX/` folder from `13_run_budget_sweep` (each run's budget read from its saved parameters): `bud_headline.csv`, `bud_value_added.csv`, `bud_convergence.csv`, `fig_bud_improvement`, `fig_bud_value_added`, `fig_bud_overtime`, `map_bud_foresight`, `bud_metadata.json`.
    - *Hazard sweep* — the main run and every complete `runs/<LC_SCENARIO>_haz<tag>/` folder from `15_run_hazard_sweep`: `haz_sweep.csv`, `fig_haz_sweep_pct`, `fig_haz_sweep_pairs`, `fig_haz_sweep_risk_box`, `haz_sweep_metadata.json`; warns if a hazard run was built from an older version of the main run's panel.

11. `11_results.qmd` — the results document. It embeds the assets from `10_results_assets` (it computes nothing itself) and adds annotation/prose for each figure, table and map. All table formatting (`flextable`) and all titles and captions live here. It reads `runs/<RUN_ID>/output_figs/` for the run set in `00_config`, so render it with `PPR_BUDGET` and `PPR_HAZARD` unset.

    Its **Updates** section serves two purposes:

    - *Sensitivity analyses* — the conversion-hazard, budget and land-cover entries (formerly the child files `11b`–`11d`). Each is shown only if the matching comparison section of `10` has run (`haz_sweep_metadata.json`, `bud_metadata.json`, `cmp_metadata.json`); otherwise it is hidden.
    - *Methods changelog* — dated, hand-written entries recording changes to the methods and the reasons for them: temporal alignment of the benefit data (7/10/2026), the fix to the land-cover spatial join (7/30/2026), the switch from net to gross habitat loss (9/15/2026), and the removal of protected areas from the hazard (9/30/2026). These entries are static text and are not regenerated by `10`; they are the place to look for the history behind the current methods.

12. `14_build_hazard_panel` — builds the inputs for one conversion-hazard run. Reads the main run's `data_panel.rds` and `eau_flows.rds`, solves for the factor `k` so that the habitat-weighted mean 2020 `trans_prob` (weights = unprotected suitable habitat) equals the `PPR_HAZARD` annual rate compounded to a decade, applies `trans_prob' = 1 - (1 - trans_prob)^k` to every row, checks the result, and writes the scaled `data_panel.rds` and `hazard_scaling.json` (target, `k`, means, source md5) to the hazard run's `derived_data/`.

Data panel

The final data panel, `runs/<LC_SCENARIO>/derived_data/data_panel.rds` (with a `.csv` copy), is written by `06` and read by `08` and `10`. Intermediate panels are written along the way: `eau_panel.rds` and `eau_flows.rds` (`03`), `panel_04_benefit.rds` (`04`) and `panel_05_risk.rds` (`05`); `14` reads `eau_flows.rds` directly. Variables in the final panel:

Variable	Type	Description
`eau_id`	integer	Unique EAU identifier. Assigned in script 01 by re-indexing valid raster cells 1:N, sorted by WMD (alphabetical) then original cell number. Stable across all scripts.
`wmd_id`	character	Name of the Wetland Management District containing the EAU centroid. 20 distinct values after exclusions.
`year`	integer	Decision time step: 2020 (baseline) and each decade 2030–2100. Nine distinct values.
`rcp`	character	Emissions scenario: `"45"`, `"85"`, `"baseline"` (2020 only), or `"stationary"` (no-change counterfactual).
`gcm`	character	Climate model variant: `"wet"`, `"dry"`, `"baseline"`, or `"stationary"`. Note that `prop_suitable` does not vary by `gcm` — only the benefit data are GCM-specific.
`prop_suitable`	double	Proportion (0–1) of the EAU classified as suitable habitat.
`abs_abundance`	double	WMD-level total duck abundance for that year and scenario, repeated across every EAU in the WMD.
`scaled_abundance`	double	EAU-level share of `abs_abundance`, allocated by habitat area. The benefit term used in the ILP.
`total_prop_suitable`	double	Sum of `prop_suitable` across all EAUs within the WMD × year × `rcp` × `gcm` group. Denominator for `habitat_share`.
`habitat_share`	double	`prop_suitable / total_prop_suitable`; the EAU's fraction of its WMD's suitable habitat in that scenario-year. `NA` where the WMD total is 0.
`trans_prob`	double	Per-period risk of habitat loss (transition probability), bounded [0, 1].
`cost`	double	Estimated acquisition cost of the EAU in nominal USD for that decision year. Scenario-invariant.
`cell_id`	integer	Raster cell number in the EAU grid — the permanent by-place key linking each EAU to its location. Used to join land-cover data to the panel by position on the ground rather than by row order. `NA` for 2020 baseline and stationary rows, which are constructed from `eau_id` after the land-cover join and never carry the cell key.

References and sources

Data. Land-cover projections: USGS FOREsce (GCAM Reference / BAU); Sohl, T. L., et al. (2012), *Agriculture, Ecosystems & Environment* 153:1–15. Waterfowl abundance: McKenna, O. P., et al. (2026), *BioScience* biag091. Acquisition cost: Nolte 2020, *PNAS* 117(47):29577 (PLACES FMV raster; Dryad doi:10.5061/dryad.np5hqbzq9). Protected areas: USGS GAP (2024), PAD-US 4.1, https://doi.org/10.5066/P96WBCHS. Conversion-rate targets for the hazard sweep: Johnston (2013), Stephens et al. (2008), Kemink et al. (2023), Rashford et al. (2011).

Model formulation. Gerling, C., M. Drechsler, K. Keuler, A. Sturm, and F. Wätzold (2023), "Time to Consider the Timing of Conservation Measures: Designing Cost-Effective Agri-Environment Schemes under Climate Change," *Agricultural and Resource Economics Review* 52(2):231–49. Wu, C.-H. (2022), *Value of flexible and dynamic resource allocation in conservation planning*, PhD dissertation, University of Melbourne.

Gurobi Optimizer Reference Manual (2026), https://docs.gurobi.com/projects/optimizer/en/current/index.html.

Other. EAU gridding adapted from a script by Heini Kujala. Prior toy-problem precedent: Rubenstein, Madeleine A., et al. "Cognitive and institutional barriers to effective natural resource management under global change." *Conservation Biology* (2026): e70393.